

Experiment 1

AIM: Develop Student Admission Management Module using OOP Concepts (Class, Object, Constructor, Access Modifiers).

Source: EXP 1.cs

Code:

```
using System;

namespace Practical1
{
    class Student
    {
        private int grNo;
        private string name;
        private string course;
        private int semester;
        private double fees;
        private bool scholarship;

        private const double ScholarshipRate = 0.10;

        // Default Constructor
        public Student()
        {
            Console.WriteLine("Student Admission Object Created\n");
        }

        // Public Properties
        public int GRNo
        {
            get { return grNo; }
            set { grNo = value; }
        }

        // Method to Accept Details
        public void AcceptDetails()
        {
            Console.Write("Enter GR Number: ");
            grNo = Convert.ToInt32(Console.ReadLine());
        }
    }
}
```

```
        Console.WriteLine("Enter Student Name: ");
        name = Console.ReadLine();

        Console.WriteLine("Enter Course: ");
        course = Console.ReadLine();

        Console.WriteLine("Enter Semester: ");
        semester = Convert.ToInt32(Console.ReadLine());

        Console.WriteLine("Enter Course Fees: ");
        fees = Convert.ToDouble(Console.ReadLine());
    }

    // Method to Calculate Fees
    public double CalculateFees()
    {
        if (fees > 50000)
        {
            scholarship = true;
            return fees - (fees * ScholarshipRate);
        }

        scholarship = false;
        return fees;
    }

    // Method to Display Details
    public void DisplayDetails()
    {
        double finalFees = CalculateFees();

        Console.WriteLine("\n==== Student Details =====");
        Console.WriteLine("GR Number      : " + grNo);
        Console.WriteLine("Name          : " + name);
        Console.WriteLine("Course        : " + course);
        Console.WriteLine("Semester      : " + semester);
        Console.WriteLine("Scholarship    : " + (scholarship ? "10% Applied" : "Not
Eligible"));
        Console.WriteLine("Final Fees     : " + finalFees);
    }
}

class Program
{
    static void Main(string[] args)
```

```
{  
    Student st1 = new Student();  
  
    st1.AcceptDetails();  
  
    st1.DisplayDetails();  
  
    Console.ReadKey();  
}  
}
```

Output:

```
Student Admission Object Created  
  
Enter GR Number: 125948  
Enter Student Name: Shrey  
Enter Course: CE  
Enter Semester: 5  
Enter Course Fees: 62000  
  
===== Student Details =====  
GR Number      : 125948  
Name           : Shrey  
Course          : CE  
Semester       : 5  
Scholarship    : 10% Applied  
Final Fees     : 55800
```

GitHub URL: <https://github.com/shrey2250/.net-practicals.git>